Improving Adversarial Robustness via Knowledge Distillation

Kaiyue Wang, Zijian Zhang, Julian Chen, Yanling Sang
kwang148@jh.edu, zzhan401@jh.edu, jchen508@jh.edu, ysang7@jh.edu

Abstract

Deep neural networks are vulnerable to adversarial attacks that introduce small perturbations and cause incorrect predictions. In this project, we study whether knowledge distillation with other method together (or with other method along) can improve the robustness of image classification models against attacks like FGSM and PGD on the MNIST dataset.

1 Introduction

Deep learning models, especially convolutional neural networks (CNNs), have achieved remarkable success in image classification tasks. On benchmark datasets such as MNIST, these models can reach very high accuracy under normal evaluation settings. However, prior work has shown that despite their strong performance on clean data, deep neural networks are highly vulnerable to adversarial examples. An adversarial example is created by adding a small perturbation to an input image that is often imperceptible to humans but can significantly change the model's prediction. This vulnerability reveals an important weakness in modern neural networks: even when the visual content of an image appears unchanged, the learned decision boundary may be unstable under carefully designed perturbations. Such a problem is especially important in safety-critical applications such as autonomous driving, medical image analysis, and security systems, where incorrect predictions may lead to serious consequences.

The goal of this project is to study adversarial robustness and evaluate whether some methods and their combinations, for example, knowledge distillation, pruning, or something else (diffusion models) can improve a model's resistance to different adversarial attacks. The input to our algorithm is a 28×28 grayscale image from the MNIST handwritten digit dataset, and the output is a digit label from 0 to 9. We first choose a baseline convolutional neural network for digit classification. Then, we will implement different methods, like use a student model to study the teacher model, or prune unnecessary weights or find a better generative model to deal with attacks. Finally, we will evaluate such methods choosen adversarial attack methods, like Fast Gradient Sign Method and Projected Gradient Descent, across different perturbation strengths. By comparing the clean accuracy and adversarial accuracy of the baseline and distilled models, we aim to understand whether knowledge distillation can improve robustness without significantly sacrificing classification performance.

2 Dataset and Features

We use the MNIST dataset (1), which contains 60,000 training images and 10,000 test images of handwritten digits across 10 classes (0–9). Each example is a grayscale image with resolution 28×28. We normalize the pixel values to the range $[0, 1]$ before feeding them into the model. Since MNIST is relatively clean and simple, we do not apply heavy data augmentation in our initial experiments. If time permits, we may consider simple augmentations such as random shifts or rotations for additional analysis.

The input images are used directly as the model input, and no handcrafted features are designed. Instead, the convolutional layers automatically learn hierarchical visual features from the raw pixels during training. This makes the project focus on the robustness of learned representations rather than feature engineering. Our evaluation will be performed on both clean test images and adversarially perturbed versions of the same images, allowing us to compare how model performance changes under attack.

3 Methods

We use a convolutional neural network (CNN) as our baseline model for handwritten digit classification. The model outputs a probability distribution over the 10 digit classes and is trained using the standard cross-entropy loss. We optimize model parameters with the Adam optimizer. This baseline provides a reference point for measuring the effect of adversarial attacks and the potential robustness improvements from knowledge distillation.

To improve robustness, we first prefer to implement the follow. We apply knowledge distillation (2) in a teacher-student framework. First, we train a teacher model on the same classification task. Then, a student model is trained using a combination of two objectives: the standard cross-entropy loss with the ground-truth labels and a distillation loss based on the softened output distribution of the teacher. A temperature parameter is used to smooth the teacher's output probabilities, and a weighting factor balances the hard-label loss and the soft-label distillation loss. We evaluate adversarial robustness using two attack methods: FGSM (3) and PGD (4). FGSM performs a single-step gradient-based perturbation, while PGD performs multiple iterative updates within a bounded perturbation region and is generally considered a stronger attack. We compare the baseline model and the distilled model under different attack strengths to study whether distillation improves robustness.

4 Deliverables

These deliverables are ordered by how essential they are to the project. We are confident that the must-accomplish goals are achievable within the project timeline.

4.1 Must accomplish

1. Train a baseline CNN model on the MNIST dataset and report its clean test accuracy.
2. Implement FGSM and PGD adversarial attacks and generate adversarial examples under different perturbation strengths.
3. Evaluate the baseline model under attack and plot accuracy-versus-epsilon curves.

4.2 Expect to accomplish

1. Implement the teacher-student knowledge distillation framework.
2. Compare the clean accuracy and adversarial robustness of the baseline model and distilled model.
3. Implement other methods to defense the attacks along or combine them with distillation framework together and report whether there exists a better method to defense a certain attack.
4. Analyze the trade-off between standard classification performance and robustness.

4.3 Would like to accomplish

1. Visualize representative adversarial examples to illustrate how small perturbations affect predictions.
2. Explore the effect of different methods and settings on the defensing attack method and see what we can improve or introduce to solve more complex attacks.
3. Extend the experiments to a more challenging dataset such as CIFAR-10.

References

References

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